

II B.Tech CSE and CS

Code : 23ACS11

Subject : OPERATING SYSTEM

UNIT-1

2-Mark Questions

1. What is an Operating System?
2. Explain different types of OS?
3. Write any five OS services?
4. What is system call? Explain the types of system call.
5. Write different operating system services.
6. What is software and its types.
7. Write all types of OS Structure.
8. What is kernel?
9. What is context switching?
10. Write difference between fork() and exec().

10- Mark Questions

1. Define an Operating System. Explain types of operating system with neat diagram.
2. Explain the operations of an operating system and explain the major functions of an operating system.
3. Discuss different computing environments in operating systems.
4. Explain various operating system services.
5. What are system calls? Explain the types of system calls.
6. Explain Failure Analysis and performance tuning.
7. Explain operating system design and implementation issues. Discuss policy vs mechanism and implementation languages.
8. Describe different operating system structures. Explain simple, monolithic, layered, microkernel, and modular systems.
9. Explain building and booting an Operating System,
10. What is operating system debugging? Explain debugging techniques used in OS debugging.

UNIT-2

2-Mark Questions

1. What is a process?
2. What are the different states of a process?
3. What is threads and its types?
4. What is a scheduler and its types?
5. Write difference between preemptive and non-preemptive?
6. What is load balancing and its types?
7. What is shared memory?
8. What is process ID?
9. What is program counter?
10. What is Starvation?

10-Mark Questions

1. Explain all process states and explain process control block(PCB) in operating system.
2. What is inter-process communication (IPC)? Explain shared memory and message passing models?
3. Define threads. Explain the types of threads?
4. Explain multilevel queue and multi level feedback queue in scheduling algorithm?
5. Explain all CPU scheduling algorithm and its advantages and disadvantages?
6. Explain Multiprocessor scheduling in operating system.
7. Consider the set of 5 processes whose arrival time and burst time are given below-If the CPU scheduling policy is FCFS, calculate the average waiting time and average turnaround time.

Process Id	P1	P2	P3	P4	P5
Arrival time	3	5	0	5	4
Burst time	4	3	2	1	3

8. Consider the set of 5 processes whose arrival time and burst time are given below-If the CPU scheduling policy is SJF non-preemptive and preemptive, calculate the average waiting time and average turnaround time.

Process Id	P1	P2	P3	P4	P5
Arrival time	3	1	4	0	2
Burst time	1	4	2	6	3

9. Consider the set of 5 processes whose arrival time and burst time are given below-If the CPU scheduling policy is Round Robin with time quantum = 2 unit, calculate the average waiting time and average turnaround time.

Process Id	P1	P2	P3	P4	P5
Arrival time	0	1	2	3	4
Burst time	5	3	1	2	3

10. Consider the set of 5 processes whose arrival time and burst time are given below-If the CPU scheduling policy is priority non-preemptive and preemptive, calculate the average waiting time and average turnaround time. (Lower number represents higher priority)

Process Id	P1	P2	P3	P4	P5
Arrival time	0	1	2	3	4
Burst time	4	3	1	5	2
Priority	2	3	4	5	5

UNIT-3

2-Mark Questions

1. List the three requirements of a solution to the critical section problem.
2. Why is Peterson's solution limited to two processes?
3. Define mutex lock.
4. Define semaphore.
5. Differentiate between binary semaphore and counting semaphore.
6. What are wait() and signal() operations?
7. What is Deadlock?
8. Write methods of deadlock?
9. What is banker's algorithm?
10. What is process synchronization?

10-Mark Questions

1. What is process synchronization? What is the important feature of critical section?
2. Explain Classical Synchronization problem
(i) Dining Philosopher problem (ii) Producer Consumer problem (iii) Reader's Writer's problem
3. Write about critical section problem, semaphores, mutex lock and monitors
4. Describe Peterson's Solution for the critical section problem.
5. Explain deadlock model and deadlock Characterization.
6. Explain the methods of handling deadlocks.
7. What do you mean by Deadlock Avoidance? Explain the use of Banker's Algorithm for Deadlock Avoidance with illustration.
8. What is Deadlock? Explain Detection and Recovery.
9. What is Deadlock? Explain Deadlock prevention & Avoidance.
10. Considering a system with five processes P_0 through P_4 and three resources of type A, B, C. Resource type A has 10 instances, B has 5 instances and type C has 7 instances. Find it is safe state or not.:

Process	Allocation	Max	Available
	A B C	A B C	A B C
P_0	0 1 0	7 5 3	3 3 2
P_1	2 0 0	3 2 2	
P_2	3 0 2	9 0 2	
P_3	2 1 1	2 2 2	
P_4	0 0 2	4 3 3	

UNIT-4

2-Mark Questions

1. What are Pages and Frames?
2. Define Dynamic Loading and Linking.
3. What are Overlays
4. Define Swapping.
5. What do you mean by Best Fit, worst fit and first bit.
6. What is Virtual memory?
7. What is HDD?
8. List all of disk scheduling algorithm .
9. What is page fault.
10. What is thrashing?

10-Mark Questions

1. Explain Contiguous memory allocation and Fragmentation.
2. Give the basic concepts about paging and techniques for structuring the page table and TLB
3. Explain the basic concepts of segmentation.
4. What is demand paging and what is its use?
5. Explain the various page replacement strategies.
6. What is thrashing and explain the methods to avoid thrashing?
7. Given five memory partitions of 100Kb, 500Kb, 200Kb, 300Kb, 600Kb (in order), how would the first-fit, best-fit, and worstfit algorithms place processes of 212 Kb, 417 Kb, 112 Kb, and 426 Kb (in order)? Which algorithm makes the most efficient use of memory?
8. A system uses 3 page frames for storing process pages in main memory. It uses the First in First out (FIFO) ,LRU and Optimal page replacement policy. Assume that all the page frames are initially empty. What is the total number of page faults that will occur while processing the page reference string given below-
 - i. 5 , 7, 6, 1, 7, 2, 1, 2, 7, 2
 - b. calculate the hit ratio and miss ratio.
9. Explain the overview of Mass Storage and explain all types disk scheduling algorithm..
10. Consider a disk queue with requests for I/O to blocks on cylinders 98, 183, 41, 122, 14, 124, 65, 67. The FCFS,SSTF,SCAN,C-SCAN,LOOK, C-LOOK scheduling algorithm is used. The head is initially at cylinder number 53. The cylinders are numbered from 0 to 199. The total head movement (in number of cylinders).

UNIT-5

2-Mark Questions

1. What is a file?
2. Mention any two file access methods.
3. What is a directory structure?
4. What is meant by file-system structure?
5. Define file-system operations.
6. What is file allocation? Name any one allocation method.
7. What is free space management?
8. What is file-system mounting?
9. What is file sharing?
10. What is an access matrix?

10-Mark Questions

1. Explain the file concept in operating systems. Describe various file access methods with suitable examples.
2. Explain different directory structures used in operating systems. Discuss their advantages and disadvantages.
3. Describe the file system interface. Explain file operations and access methods supported by operating systems.
4. Explain the file-system structure and describe the steps involved in file-system implementation.
5. Describe various techniques used for free space management in file systems.
6. Explain the concept of file sharing in operating systems. Discuss consistency and access control issues.
7. Explain file-system mounting. Describe partitions and mounting with suitable diagrams.
8. Explain the goals and principles of protection in operating systems.
9. Explain protection rings and the domain of protection with suitable examples.
10. Explain the access matrix model for protection. Discuss how it can be implemented in operating systems.